MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER B.Sc(Hons) DEGREE EXAMINATION
MATHEMATICS(DSC)
(2024 ADMISION ONWARDS)
MG4DSCMAT201-FUNDAMENTALS OF ANALYSIS

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Exhibit a bijection between $\mathbb{N}$ and a proper subset of itself. [U][1]
2. There does not exist a rational number r such that $r^2 = 2$. [A][2]
3. State the completeness property of $\mathbb{R}$. [K][3]
4. If $S = \{\frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N}\}$, find inf S and sup S. [An][3]
5. Show that $Re(iz) = -ImZ$. [U][5]
6. Define accumulation point of a set. [U][6]
7. Find $Log(-1)$. [U][8]
8. Find $|e^{2+2\pi i}|$. [A][8]

Part B
Answer any five questions. Each question carries 6 marks

9. Prove in detail that if S and T are denumerable, then $S \cup T$ is denumerable. [U][1]

10. Prove that if $a \in \mathbb{R}$ is such that $0 < a < \epsilon$ for every $\epsilon > 0$, then $a = 0$. [A][2]

11. State and Prove the Archimedean Property of $\mathbb{R}$. [An][3]

12. State and Prove nested interval property. [An][4]

13. Find the $\text{Arg}(-1 - i)$. [U][5]

14. Prove that $f(z) = z^2$ is differentiable everywhere. [A][6]

15. Prove that $log(z_1 z_2) = log\ z_1 + log\ z_2$. [A][8]

16. Show that $sin^2 z + cos^2 z - 1 = 0$. [A][8]

Part C
Answer any three questions. Each question carries 10 marks

17. (a) Determine the set $A = \{x \in \mathbb{R} : |x - 1| < |x|\}$. [A][2]

    (b) Solve the inequality $|2x - 1| \leq x + 1$. [A][2]

18. Let $S = \{x \in \mathbb{R} : x > 0\}$. Does S have lower bounds? Does S have upper bounds? Does inf S exist? Does sup S exist? Prove your statements. [An][3]

19. Prove that the set $\mathbb{R}$ of real numbers is not countable. [An][4]

20. Show that $u(x, y) = y^3 - 3x^2 y$ is harmonic throughout the entire plane and find its harmonic conjugate. [A][7]

21. (a) Show that $sinh\ (z + \pi i) = -sinh\ z$. [A][8]

    (b) Show that $cosh\ (z + \pi i) = cosh\ z$. [A][8]

22. Prove that $tan^{-1} z = \frac{i}{2} log \frac{i+z}{i-z}$. [An][8]